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Plagiarism Detection in AI Generated Content

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ABSTRACT—

The extremely fast development of generative Artificial Intelligence (AI) and text "humanizer" tools is a serious problem for the maintenance of intellectual and academic honesty since these systems very often produce content that is very similar to human writing. This study aims at creating a better method for detecting plagiarism by clearly differentiating the quantifiable writing patterns of human and AI-generated texts. The main difference is that AI-generated texts are, in general, deprived of the nuances, spontaneous variations, complexity, and authenticity that are typical of human expressions. Our proposed approach is looking at a deep manner to utilizing and tuning advanced Natural Language Processing (NLP) models, e.g. BERT (Bidirectional Encoder Representations from Transformers).

The method is concentrated on the extraction of the main linguistic and statistical features:

Perplexity: This is a metric for determining the unpredictability of the word sequences, where a lower value shows that the text is more predictable and is thus AI-generated.

Burstiness: This metric is used for gauging the extent of natural variation in the lengths of the sentences, their grammatical structures and rhythms, and it is usually very high in human texts whereas very low in the uniformly structured AI output.

The model is tuned using a labeled corpus of human and AI-generated texts to decide which type the given content belongs to based on these subtle differences. The findings signify the excellent capabilities of the adjusted model in locating the text produced by a machine (e.g., the model can be up to 95% accurate in a multi-model approach), hence, opening the route for a scalable and efficient framework to confront AI plagiarism problem.

I. Introduction

The emergence of advanced generative AI models like ChatGPT and GPT-4 has, to a great extent, changed the way new content is created in the academic, professional, and creative sectors. While such a technological breakthrough is capable of raising productivity levels, it also puts originality and academic integrity at risk. To compete with the capabilities of current plagiarism detection tools, AI systems even those that are specifically designed to "humanize" their output, are becoming increasingly sophisticated in creating texts.

One of the main weaknesses of earlier plagiarism detection tools is that they depend on simple lexical similarity (string-matching) and, therefore, cannot detect content that has been contextually rewritten or paraphrased by AI models.

This research highlights the urgent demand for detection instruments of a new generation. The foremost aim is to locate stylistic features that can distinguish human writing from AI writing without resorting to mere pattern matching. As a result of the subtle differences in linguistic structure and complexity which the generative model still cannot replicate fully, this study intends to create a strong, fast, and efficient solution that is able to identify real human authorship apart from machine generation.

II. Related Work

First attempts at plagiarism detection heavily relied on rule-based systems and basic text-matching techniques. These techniques were rendered almost completely inefficient with the emergence of AI and paraphrasing technologies, since the latter capable of changing the text that they still carry the same semantic meaning are not detected.

A. *Early Machine Learning Approaches*

Trying to spot AI-generated texts via Machine Learning (ML) models like Deep Learning and Extremely Randomized Trees Classifier was a good idea in principle, but the results were not always reproducible, the models were not robust, and they exhibited a high rate of false positives. For example, one research on GPT-4 detection showed that the method used was on average only 67% accurate and that false positives were as high as 45%, i.e., almost every other human text was recognized as AI-generated. In fact, these drawbacks and shortcomings led the authors to acknowledge that more sophisticated and subtle models are needed.

B. Feature-Based and Transformer Models

The domain has evolved to depend on models that have the capacity of uncovering more complicated linguistic features.

Statistical and Linguistic Features: The researchers came up with a number of distinct and measurable features, namely:

Burstiness and Perplexity: These were the metrics that the tested early detection systems (such as one based on GPT-2) largely used to differentiate the texts from one another. Here the main idea is that human-produced text is characterized by having high burstiness (variation in sentence structure) and high perplexity (the unpredictability of the word sequence), on the other hand, artificially created text is rather uniform and highly predictable (low perplexity and low burstiness).

Transformer Models (BERT/RoBERTa): The introduction of the Transformer-based architecture in the form of BERT (Bidirectional Encoder Representations from Transformers) and other models proved to be a substantial leap in AI-generated content detection. These models, which are trained on enormous datasets, have the ability to spot subtle, complex linguistic as well as logical features even in the shortest texts to a much higher degree than other models. Being able to effectively tackle problems such as generalization and dataset imbalance, BERT is considered by far the most accurate method of AI-generated content detection. The performance of fine-tuned BERT-based models has been demonstrated to be better than such as RoBERTa and DistilBERT.

Hybrid Systems: The greatest success in the field of AI-generated text detection has been brought about by the concerted effort of several differently specialized transformer models, e.g. multitask-qa-mpnet-base-dot-v1 for semantic similarity along with a detector model like roberta-large-openai-detector specifically for pinpointing the AI-generated content. This synergistic approach can be very effective in ensuring that the highest accuracy level in real-time detection is attained.

III. Detailed System Architecture

This system design makes use of a layered approach that not only clearly differentiates human from AI writing but also goes beyond mere classification by involving deep contextual analysis along with statistical feature extraction..

A. Input and Preprocessing Pipeline

The system is built to take the raw text that is to be analyzed. It is necessary that the data preprocessing operations should be done in order to maintain the quality of data and to make the model work effectively:

Normalization: It is the part of text cleaning when urls, acronyms, and words that are repeated are removed.

Noise Removal: Standardizing texts by making all characters lowercase and removing stop words.

Tokenization: The BERT specialized tokenizer is used to convert the text to the format that the transformer model can accept.

B. Core Detection Module

The main part of the system is a fine-tuned, BERT-based, uncased model that has been optimized for the classification of texts. Besides, a robust system can have a set of different transformer models where the AI detector component (e.g. roberta-large-openai-detector) is used for solving the content authenticity problem.

C. Feature Extraction and Quantifiable Differentiation

The system is not limited to using the deep contextual abilities of the transformer only, but it also calculates a number of important key statistical indicators so that it can merge them and thus exploit the predictability of AI patterns to the full extent:

Perplexity Score: The purpose of a perplexity score is to evaluate how predictable a sequence of words is. When the perplexity score is really low, the corresponding source of text is most likely to be a non-human one.

Burstiness Score: The calculated score helps to describe the variance and the diversity of the sentence structure, length, and tempo. A lower score is a sign of the uniform, repetitive structure that is typical of AI-generated text.

D. Classification and Output

The final layer is a SoftMax classification function that determines the probability of the input text to be a member of one of the two classes Human-Generated (0.0) or AI-Generated (1.0). The decision about the final classification is made by setting a threshold on these probabilities. Apart from that, the system can also output an analytical report that stands for the thorough content analysis and comprises the AI likelihood score as well.

IV. Proposed Methodology

The methodology that we plan to develop and intelligently equip the AI plagiarism detection model with is an orderly manner that mainly focuses on those linguistic differences that are very close but still different between human and machine texts.

A. Data Collection and Preparation

We require a very large volume of labeled data, roughly 500,000 examples. The data should fairly represent the two types of texts, i.e., human-written and AI-generated, so that the model does not become biased towards one of them. This collection of texts is necessary for the model so that it can be trained to find the most genuine human authorship features.

B. Preprocessing and Chunking

Initially, the preprocessing steps referred to in the diagram are performed on the raw data. In order to overcome the limitations of the computer and memory when training the dataset, it is properly divided, and each division or chunk is processed individually.

C. Feature-Based Detection Mechanism

The fundamental point of the human vs. AI patterns comparison is feature per feature:

Perplexity Calculation: The language model is used for determination of the perplexity of the input text. If the perplexity is high, it indicates that the text is quite unexpected in terms of the use of varied words and complicated grammatical structures, thus, it is actually a feature of human writing.

Burstiness Calculation: Statistical methods are brought in for locating the differences in sentence structure and flow. If a high variance (burstiness) is found, it is possible to use this as an indication that the text is most probably a naturally written one, not by a computer program.

D. Model Fine-Tuning and Optimization

The selected pre-trained transformer model (e.g., BERT) is fine-tuned for performing binary text classification task:

Loss Function: When it comes to the classification task, the loss function which is used is sparse categorical cross-entropy loss.

Optimizer: The Adam optimizer is used in conjunction with the backpropagation algorithm to make the changes in model weights in the most efficient manner.

Hyperparameter Tuning: The main reason for the optimal hyperparameter setting is to get a good balance amongst accuracy, precision, and recall and, hence, it is very important for the model to be able to perform at a high level.

Such fine-tuning of the model aims at uncovering those human language intricacies the model might not see and which basically is the way the model differentiates the AI output.

V. Experimental Setup

We want to verify our detection method through detailed comparative experiments with the chosen baseline models. The presented experimental setup is aimed at a profound and unbiased assessment of the system's potential.

A. Dataset: The Basis for Training and Testing

Quality and properly annotated text datasets are the essential conditions for correct training and reliable evaluation.

Source and Labeling

As a source of content for AI detection research, we will use large and already labeled datasets such as those from Kaggle that are suitable for the task. Each piece of text will be unequivocally labeled as one either human-written (0.0) or AI-generated (1.0), which is a necessity for any supervised classification task.

Scale and Balance

The dataset is going to contain nearly 500,000 text instances, the number of human and AI-generated texts will be the same so that the model does not overly fit or become biased. Such a balanced and large dataset is a guarantee that the model will be able to generalize to both classes.

Data Splitting

Your data will be divided into three different groups:

Training Set – A dataset for the model to learn from. **Validation Set –** Parameter tuning capability and overfitting check during training.

Test Set – A completely different, non-overlapping subset that is used for the final evaluation, thus providing a fair test of model performance.

B. Models and Baselines: The Comparative Arena

We can demonstrate the effectiveness of our method by the first and foremost experiments that compare our suggested models' performance with the baseline methods.

Proposed Model Architectures Fine-Tuned BERT (Base Model)

Our main discriminator will be a locally fine-tuned version of the BERT (uncased) model. Only the final layers of the network being updated for a binary classification task, the model is specifically adapted to the human vs. machine text classification.

Composite Transformer Ensemble (Advanced Option)

We want to realize the next-level method by the ensemble that combines the results from several transformer networks (e.g., models like roberta-large-openai-detector, etc.). The diversity of the models can result in a more accurate and stable output.

Baseline Models for Comparison

Reasoning where our discoveries fit into the landscape of what has been known so far, we are going to put side by side the performance of our system with that of these methods:

RoBERTa

The modifications that resulted in RoBERTa turned it into a strong transformer baseline, which has been demonstrated to be better than the original BERT in many NLP tasks, among others.

DistilBERT

Another model from the BERT family, but this time a smaller and more efficient version of BERT that is capable of showing what one has to give up when selecting efficiency over performance.

ExtraTrees Classifier

An old, ensemble-based machine learning method of the non-deep-learning type. Its position is that of a baseline, checking how far the "traditional" methodology can go before the transformer architectures, which are the state-of-the-art, start to demonstrate their performance advantages.

C. Evaluation Metrics: Quantifying Performance

Four standard classification metrics that are listed below will be the measures of the performance of the classifiers to be put to use.

Accuracy

This metric demonstrates the number of times the model is right out of all the attempts made by it.

Precision

If we consider the AI-generated texts, precision tells us how many of those, which were identified as such, were actually AI-generated ones? This is very important when the number of false positives has to be kept at a minimum.

Recall

How many of the AI-generated texts did the model manage to find if we consider all of them? A high recall means low false negatives in this case.

F1-Score

1-score is the harmonic average of precision and recall. It shows in one number the balance between recognizing AI-generated content and committing false positives.

VI. Results and Discussion

A. Performance Against Baselines

The experimental result of the proposed fine-tuned

Precision: 80%

Recall: 60%

F1-Score: 69%

Moreover, multi-model configurations that employ advanced transformer models for different tasks have resulted in a considerably high detection rate of more than 95% accuracy. Consequently, traditional string-matching and lexical methods were largely left behind in terms of performance.

B. Analysis of Differentiating Features

The augment in performance can be taken as a proof of the main idea that there are different linguistic features which can be used to distinguish between human and machine text.

Failure of Earlier Models: Previous black-box ML models that were used in the research were very unclear and thus prone to misclassification, as the study found that almost half of genuinely human text was falsely identified as AI-generated (45% false positives). That is why the models did not have enough subtlety to correctly handle the variability of humans.

Success of Transformer Models: The fine-tuned BERT model, in particular, demonstrated this by its ability to identify complex syntactic and semantic features in machine-generated texts is the most significant one.

Help of Statistical Features: The metrics that have been implemented successfully such as Perplexity and Burstiness were of great importance. The model leveraged the higher-order predictability and uniformity of AI output (low perplexity and low burstiness) to distinguish it from the irregularity of human language.

The results establish that a carefully-designed system that combines state-of-the-art transformer analysis with the recognition of natural stylistic features is the most effective way to pinpoint machine-generated texts and preserve the authenticity of the content.

VII. Limitations and Future Work

Limitations

"Humanizer" Evasion: One of the problems with generative AI is that the technologies aimed at "humanizing" the generated output are constantly improving. Therefore, the difference between a text written by a human and a machine is becoming smaller and smaller. That is why the detection models need to be updated continuously.

BERT-based model has been excellent by any single metric of False Positives: Although the performance has been improved, the the evaluation in comparison to the baseliner models such as necessity to lessen the number of false positives is still the main

RoBERTa and DistilBERT. To be specific, the fine-tuned BERT limiting factor. The problem of mistakenly identifying a genuine

model has led to the following results: Accuracy: 72%

human-written text as an AI-generated one is serious both from the academic and ethical point of view.

Generalization: The earlier machine learning models had difficulties with generalization and dataset imbalance, which is still an issue that needs to be constantly addressed by employing more robust pre-training and fine-tuning strategies.

Future Work

Continuous Framework Updates: The scalable framework, as proposed, should not only be efficient and robust but also be able to accommodate regular updates that are in line with the latest generative AI and "humanizer" technology developments.

Multimodal Detection: The scope of the research should not only be limited to text but also be extended to develop the systems capable of detecting AI-generated plagiarism in multimodal content such as images, audio, and video.

Enhanced Nuance and Interpretability: The future work should be oriented towards the development of more nuanced solutions for dealing with misclassifications and enhancing model interpretability so that detection decisions are clear and trustworthy.

Conclusion

This research has successfully been a venture in figuring out a robust plagiarism identification technique based on a transformer model, a system that understands the essential differentiation between human and AI writing styles. The differences in linguistic complexity, syntactic variation, and statistical features like perplexity and burstiness were used to create a very effective classification system. The refined transformer model demonstrated increased capability, thus, it was the main reason to argue that the machine-generated text's inherent predictability and uniformity still constitute exploitable weak points. Such a scalable structure is a feasible and cost-effective route to keeping one step ahead of the problems arising from the use of generative AI and humanizers, thus fulfilling the condition that the detection method needs to get better for a safe intellectual and academic integrity environment.

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